VoiceOps Report

VoiceOps 2026 – ZentroMart Support Agent: Report

What was built

1. **Step 1 – Prompt-only baseline.** LLM (Gemini free tier, OpenRouter free fallback) with the support rules in the prompt, no KB.
2. **Step 2 – RAG.** Section-level chunks with [SECTION-ID], BM25 + Hindi/Hinglish query expansion, exact SKU/pincode matching, product→policy hop, precedence weighting (policy > guide > help > FAQ > marketing; superseded down-weighted). Replies end with the cited chunk IDs.
3. **Step 3 – Multi-turn agent.** Tool loop (lookup_order, create_ticket, date_calc, search_docs). Enforced in code: caller verification before order data, lock after 2 failed attempts, verification values scrubbed from tickets, "ticket created" only after a real ticket exists.
4. **Step 4 – Voice.** Gnani STT in, Gnani TTS out. Replies are rewritten for the ear (amounts, dates, IDs spoken). Batch command: `python run.py --audio-dir DIR --out sample_outputs`.

Results (first 30 public questions + 10 conversations / 34 turns)

Measure	Step 1 prompt-only	Step 2 RAG
Answer pass rate (must-include / must-not)	26.7%	80.0%
Decision correct (answer/clarify/escalate)	90.0%	86.7%
Retrieval recall vs gold evidence	n/a	87%

Step 3 conversations (RAG + tools): turn pass 85.3%, text 91.2%, correct tool use 94.1%.

RAG tripled the pass rate: without the KB the model cannot know policy numbers, windows and fees, so it guesses.

What went wrong

- **Free-tier limits.** OpenRouter free cap (50/day) killed the first run; switched to Gemini free tier with resumable scoring. Because of this, only the first 30 questions were scored, not the full set.
- **Conversation retrieval.** Recall is lower in multi-turn (≈41%) because short follow-ups (“and for that one?”) carry little query text.
- **Clarify vs answer.** The agent sometimes answers when it should ask for the order ID/verification first (e.g. PC005, PC007, PC009, PC010 early turns), and sometimes escalates late.
- **Heuristic scoring.** Pass/decision checks are keyword-based, so some false fails/passes are possible.
- **Voice.** Very quiet or short mic clips gave empty transcripts; added gain normalisation and a “didn’t catch that” reply. Live voice quality was not scored.

Would fix next

Rewrite follow-up queries with history before retrieval; add a pre-answer “need verification?” check; run the full eval with a paid or higher-quota model; score ASR-noisy inputs.